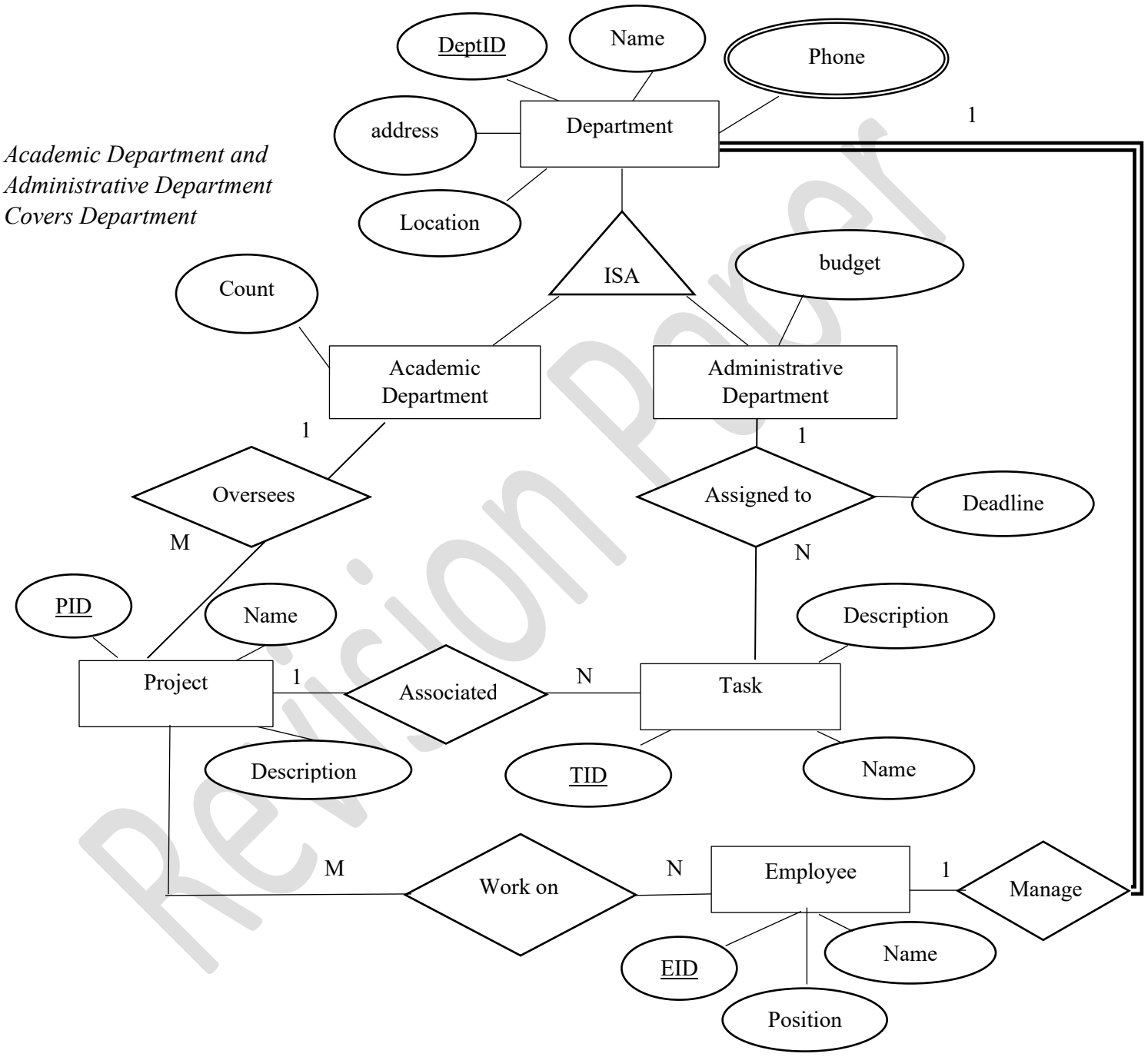
Consider the following EER Model.

Academic Department and Administrative Department Covers Department



Revision Paper

1. What is the best option for mapping the ISA hierarchies in the diagram?

    a) Option 1
    b) Option 2
    c) Option 3
    d) Option 4
    e) Option 5

2. Based on the EER diagram, which of the following is the most accurate description of the cardinality between Department and Employee in the Manage relationship?

    a) One department manages exactly one employee.
    b) Each department can manage many employees, but each employee is managed by one department.
    c) Many departments can manage one employee.
    d) Employees manage departments.
    e) Department must be managed by an employee. Employees may or may not manage departments.

Consider a relation **R (ProjectNo, ProjectName, EmpNo, EmpName, DeptNo, DeptName, HrsWork)** with the following set of functional dependencies F over R:

F = { ProjectNo->ProjectName

DeptNo->DeptName

EmpNo->EmpName, DeptNo, DeptName, HrsWork}

3. Find all the keys in relation to R

    a) ProjectNo
    b) DeptNo
    c) EmpNo
    d) EmpNo+ProjectNo
    e) EmpNo+DeptNo

Revision Paper

4. Which normal form is **R** in ?

   a) 3NF
   b) BCNF
   c) 1NF
   d) 2NF
   e) UNF

5. Which of the following represents the correct final set of relations after normalization?

   a) R1 (ProjectNO, ProjectName)
   R2 (EmpNo, EmpName, DeptNo, DeptName, HrsWork)
   R^1 (ProjectNO, EmpNo, HrsWork)

   b) R1 (ProjectNO, ProjectName)
   R2 (DeptNo, DeptName)
   R^1 (ProjectNO, EmpNo, HrsWork)

   c) R1 (ProjectNO, ProjectName)
   $R2^1$ (EmpNo, EmpName, DeptNo, HrsWork)
   R3 (DeptNo, DeptName)
   R^1 (ProjectNO, EmpNo)

   d) R1 (ProjectNO, ProjectName)
   R2 (EmpNo, EmpName, DeptNo, DeptName, HrsWork)
   R3 (DeptNo, DeptName)

   e) R1 (EmpNo, EmpName, DeptName, HrsWork)
   R2 (DeptNo, DeptName)
   R^1 (ProjectNO, EmpNo, HrsWork)

Revision Paper

Structured Questions

Consider the following schema of a database designed for a Library:

> **Book** (bookId: int, title: varchar(100), author: varchar(100), isbn: varchar(20), publicationYear: int, genre: varchar(50), availableCopies: int)
>
> **Member** (memberId: int, firstName: varchar(50), lastName: varchar(50), email: varchar(50), phone: int, address: varchar(100))
>
> **Loan** (loanId: int, bookId: int, memberId: int, loanDate: date, dueDate: date, returnDate: date)
>
> Fine (fineId: int, memberId: int, amount: real, paymentStatus: varchar(50))
>
> **Librarian** (librarianId: int, name: varchar(50), email: varchar(50), phone: int, address: varchar(100))

The **'Book'** table stores information about books available in the library, including unique book ID, title, author, ISBN, publication year, genre, and the number of available copies. The **'Member'** table holds details about library members, such as their unique member ID, first name, last name, email, phone number, and address. The **'Loan'** table manages book loans, with each loan having a unique ID and being associated with a specific book and member. It records the loan date, due date, and return date. The **'Fine'** table stores information about fines incurred by members for late returns or other penalties. It includes a fine ID, member ID, fine amount, and payment status. The **'Librarian'** table contains details about library staff, with each librarian having a unique ID, name, email, phone number, and address.

a) Write SQL Queries to perform the following:
   i. Find the member's name and phone number who borrowed a book with the title 'The Great Gatsby'.
   ii. Find the member who has the highest total amount of fines than all the other total fine amounts. Find the member's id and address.
   iii. Find the book titles, authors, and member names of books currently on loan.

b) Create a function to calculate the total fine amount for a given member. This function should account for fines with a late penalty of 10% applied to those marked as 'overdue' payment status.

c) Create a trigger that automatically updates the 'TotalFineAmount' column in the 'Members' table whenever a new fine is added, or an existing fine is updated. The 'TotalFineAmount' column should contain the total fine amount for each member. Ensure the trigger adjusts the 'TotalFineAmount' column accurately to reflect any partial payments made by members. The member pays only 50% of the total fine amount as an initial payment when the payment status is marked as 'Partial'.

Revision Paper